

Intuitive Gesture-Based Interaction with 4D Objects in Virtual Reality

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ABSTRACT

We present a gesture-based VR system that enables intuitive manipulation of four-dimensional (4D) objects and point clouds through a three-dimensional perspective view in immersive virtual space. We design a set of hand gestures for basic 4D operations, including translation, scaling, and two types of rotations. One of the rotation gestures specifically supports 4D rotations involving an invisible axis that cannot be directly perceived in the 3D projection. By immersing users in the virtual environment, the proposed system allows them to directly manipulate and deform 4D objects using natural hand gestures, demonstrating the feasibility of intuitive interaction with higher-dimensional data.

Index terms: 4D space, 4D objects, virtual reality, hand gesture, interaction, basic operations, rotation.

1 INTRODUCTION

High-dimensional spaces, shapes, and data have appeared in many fields, including geometry, data base, and robotics. With recent advances in big data, cloud computing, and artificial intelligence, processing and visualization of high-dimensional data are expected to become increasingly important challenges. High-dimensional objects and data are not only targets of symbolic computation, but have also been explored as targets of intuitive human interaction. For example, Matsumoto et al. [1] proposed a VR system that visualizes 4D objects by projecting them onto 3D subspaces and allows users to manipulate them using a controller for head-mounted display (HMD). Bosch [2] proposed an interactive system that visualizes rigid bodies in an N-dimensional space using 3D cross-sections and supports basic operations as well as collision detection between objects. However, no prior system has allowed users to fully immerse themselves in a VR environment and manipulate 4D objects or point clouds using intuitive gestures.

In this study, aiming at enabling users to immerse themselves in multidimensional spaces and interact with multidimensional objects and points, we present a system that enables intuitive manipulation of 4D objects and points within a VR environment obtained by projecting 4D space into the 3D perspective space. In this system, intuitive hand gestures—designed to respect the visual imagery associated with each operation—are defined for basic operations such as translation, scaling, and two types of rotations (ordinary one and specific one relating to the invisible fourth axis). Furthermore, using 4D objects such as regular polytopes, as well as 4D points, we confirmed that manipulation via gestures is practically feasible.

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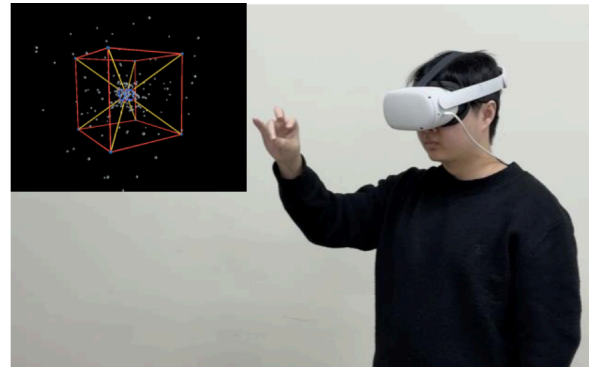


Figure 1: 4D rotation of a hypercube with multiple points.

2 VISUALIZATION AND GESTURES

2.1 Visualization

The four-dimensional space is treated as a Euclidean space with four orthogonal axes (x, y, z, w), and its representation is obtained by perspective projection onto the xyz -space at $w = 0$. The projected image is assumed to be displayed as a virtual environment in a head-mounted display (Meta Quest 2). In this paper, we focus on rendering 4D objects and points in this view, as well as enabling basic 4D operations via hand gestures.

2.2 Gestures for Basic Operations

Translation: By touching the pads of the right thumb and index finger together — as if pinching the object — as shown in Fig. 2(a) at the target position and then moving the hand, the object is translated in the same direction as the hand motion.

Uniaxial scaling: By touching the pads of the right thumb and index finger on both hands and bringing the hands closer together, the object is compressed along a single axis; moving the hands apart produces expansion along that axis as shown in Fig. 2(b).

Scaling: By touching the right thumb and middle finger together at the target object and pushing the hand forward (away from the body) as shown in Fig. 2(c), the user enlarges the object; releasing the fingers stops the effect. Pulling the hand backward (toward the body) makes the object shrink.

Rotation: We define gestures for two types of rotations.

(1) The rotation within the xyz -space is defined by touching together the pads of the thumb, index finger, and middle finger of the left hand, and performs a rotating motion on the plane with the desired direction as shown in Fig. 3(a).

(2) For the rotations whose invariant plane includes the w -axis, the gesture is defined by touching the pads of the right thumb and ring finger, moves the hand backward toward the body, and finally releases the fingers as shown in Fig. 3(b). The reverse rotation is obtained by moving the hand in the opposite direction. With this gesture, rotations of 4D objects such as a hypercube become possible in which in the 3D perspective view, the outer cube deforms, shrinks inward to become an inner cube, and then the cubes appear to exchange with each other, which resembles the defined gesture itself.

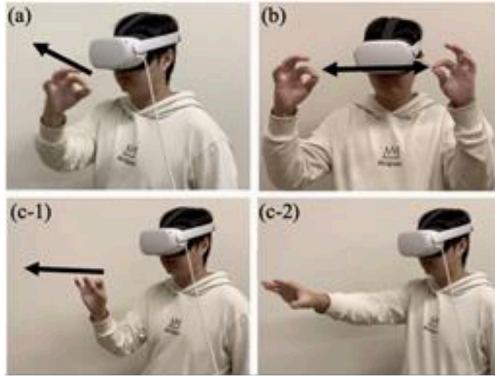


Figure 2: Gestures for translation and scalings.

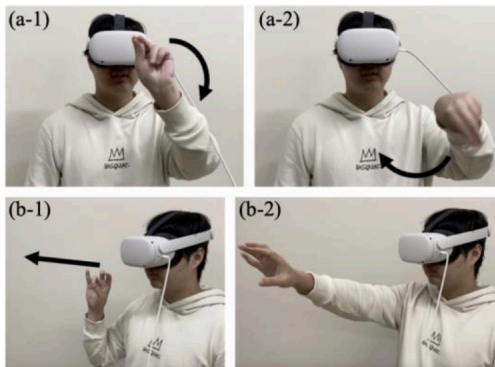


Figure 3: Gestures for two types of rotations in 4D.

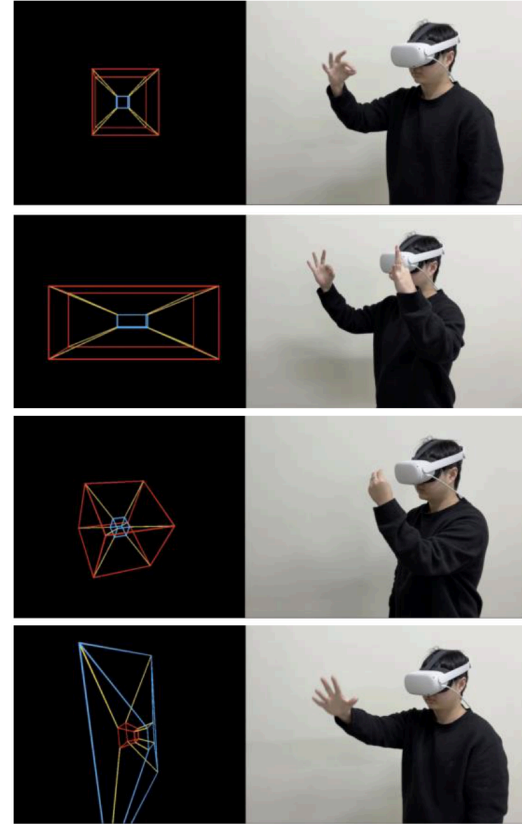


Figure 4: (a)-(d) Demonstration of basic operations in 4D.

3 IMPLEMENTATION

The system was implemented in C# using the Unity (version 2022.3.20f1c1). For hand tracking, we used the four outward-facing cameras mounted on Meta Quest 2 headset and Unity's XR Hands package and recognized each gesture defined in Section 2 by using 24 hand joints, including one point on the wrist.

4 DEMONSTRATION

We examined how 4D objects could be manipulated in VR as projected representations using the proposed gestures, across various shapes and point clouds including 4D polytopes. Fig. 4 illustrates some of the resulting images of the operations on the 4D hypercube along with the gestures; (a) translation; (b) uniaxial scaling; (c) rotation in 3D; and (d) rotation with invariant planes including the w -axis.

We also illustrate one more example of manipulating 4D hypercube and hyperrectangular parallelepiped with 200 points inside and outside them in Fig. 5, which demonstrates how we can immerse ourselves in the virtual space and intuitively manipulate the shapes and points that originates in the 4D space.

5 CONCLUDING REMARKS

As a first step toward intuitive interaction with multidimensional data in VR, we proposed a method for manipulating shapes and points in 4D space with intuitive hand gestures and verified the basic functionality. During the conference demo, attendees will be able to try the system and discuss potential applications.

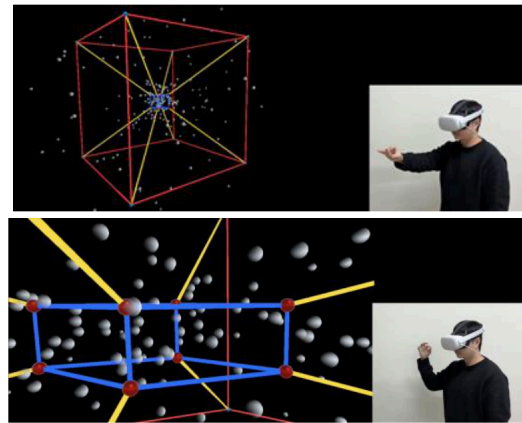


Figure 5: Manipulation of 4D shapes and points.

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